

Info Stealer Detection & Response

Integrated SIEM & SOAR Project

Information Security Analyst — Project Report

1. Introduction

Idea

The core concept is to emulate an Information Stealer malware in a controlled environment, capture its behavioral logs, and build an Semi - automated Security Operations Center (SOC) pipeline. This involves detecting the threat via a SIEM solution, forwarding the alerts to a SOAR platform, and automating the Incident Response (IR) steps for containment and eradication.

How It Is Applied

- Executing the Info Stealer safely to generate telemetry.
- Ingesting logs into Elastic Security (SIEM).
- Creating custom detection rules to detect Info Stealer.
- Utilizing a custom Python service to pull alerts and push them to a Shuffle SOAR Webhook.
- Automating threat intelligence gathering and alerting through SOAR playbooks.

How It Helps in Cybersecurity Defense

It drastically reduces the Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR) That affects the SLA. By automating Tier 1 SOC analyst tasks (like IP reputation checking and alerting), the defense team can focus on complex threat hunting rather than manual triage.

Add-Ons

Threat Intelligence enrichment using the VirusTotal API v3 integrated directly into the automated playbook.

Future Additions

Integrating active containment mechanisms, such as pushing API commands to edge firewalls or EDRs (like Windows Defender/CrowdStrike) to automatically isolate the infected host from the network without human intervention.

2. Methodology

2.1 Technology Stack

Tool	Role	Function
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Elasticsearch	SIEM — Data store	Indexes and searches all ingested log events.
Kibana	SIEM — UI & rules	Detection rules, alert management, dashboards.
Filebeat	Log shipper	Collects endpoint logs and forwards to Elasticsearch.
Shuffle	SOAR engine	Executes response playbooks triggered by Kibana alerts.
Scripting: Python 3	SIEM	using requests, datetime libraries for API integration

2.2 Timeline

Phase	Activity	Period
Setup	Environment design & tool installation	Febrarury
Ingestion	Log pipeline configuration & data validation	March
Detection	Kibana rule development & dashboard creation	March
Automation	Shuffle integration & playbook development	April
Testing	End-to-end simulation with Atomic Red Team	April - May
Docs	Report writing & final review	May

2.3 Implementation

The idea was applied by configuring a continuous background service (elastic_to_shuffle.py) that queries the internal.alerts-security.alerts index in Elastic every 60 seconds. Upon detecting a signal event, the script extracts the raw JSON payload and POSTs it to a Shuffle Webhook. Shuffle then extracts the malicious IP, queries VirusTotal, and uses conditional logic (malicious > 0) to trigger an automated Email alert containing the extracted IOCs.

3. Proposed Impact

Objectives

- Successfully simulating and detecting advanced malware behavior.
- Establishing a zero-touch alert forwarding mechanism.
- Building a functional OSINT automation inside a SOAR platform.

Percentage of Achievement

~85% to 90% (The detection, ingestion, and enrichment pipelines are fully operational. The remaining percentage accounts for the final physical eradication/isolation scripts).

Difficulties Faced

- Handling API pagination and authentication headers.
- Formatting complex JSON payloads to be accepted by webhooks.
- Resolving dynamic data mapping errors inside the SOAR platform (e.g., translating raw variables like \$exec.source.ip into actual IP addresses for third-party APIs).
- Configuring cloud vs. local API keys for SMTP/Email routing.

The Future Adds

Developing automated eradication playbooks that delete the malicious payload hashes from the endpoint via EDR integration.

4. Results

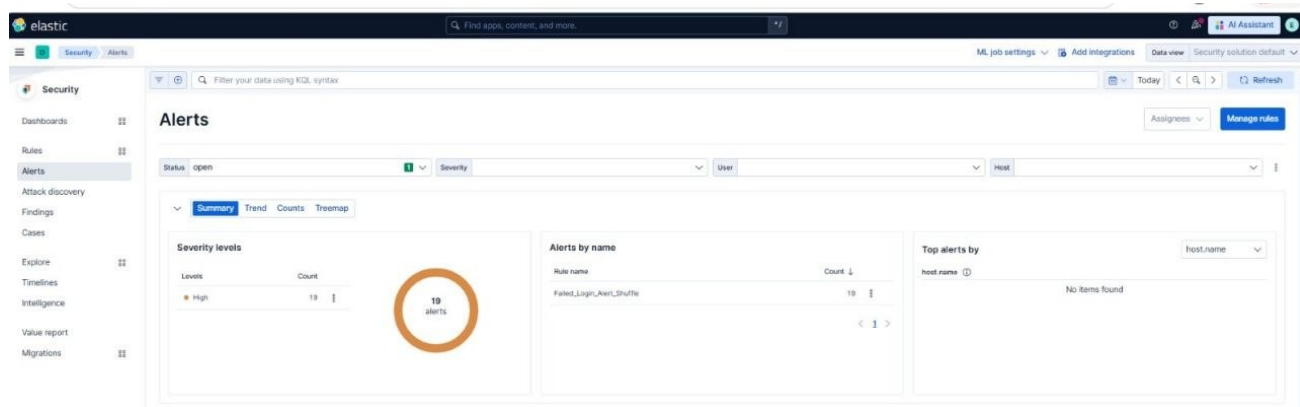


Figure 1: Elastic SIEM showing the triggered alert

```
[{"_index": "filebeat-test", "_id": "OKH_kJ0BX0Su9_ZuJ0-F", "_version": 1, "result": "created", "_shards": {"total": 2, "successful": 1, "unsuccessful": 1}}]
Shuffle$ journalctl -u shuffle_alert.service -f
Apr 15 13:46:01 DESKTOP-72C1QPJ python3[1117]: [2026-04-15 13:46:01.325218] No new SIEM alerts found.
Apr 15 13:47:01 DESKTOP-72C1QPJ python3[1117]: [2026-04-15 13:47:01.376683] No new SIEM alerts found.
Apr 15 13:48:01 DESKTOP-72C1QPJ python3[1117]: [2026-04-15 13:48:01.417244] Found 1 SIEM Alerts. Sending to Shuffle...
Apr 15 13:49:01 DESKTOP-72C1QPJ python3[1117]: [2026-04-15 13:49:01.537995] Found 1 SIEM Alerts. Sending to Shuffle...
Apr 15 13:50:01 DESKTOP-72C1QPJ python3[1117]: [2026-04-15 13:50:01.684139] Found 5 SIEM Alerts. Sending to Shuffle...
Apr 15 13:51:02 DESKTOP-72C1QPJ python3[1117]: [2026-04-15 13:51:02.228633] Found 5 SIEM Alerts. Sending to Shuffle...
Apr 15 13:52:02 DESKTOP-72C1QPJ python3[1117]: [2026-04-15 13:52:02.821308] Found 6 SIEM Alerts. Sending to Shuffle...
Apr 15 13:53:03 DESKTOP-72C1QPJ python3[1117]: [2026-04-15 13:53:03.972758] Found 5 SIEM Alerts. Sending to Shuffle...
Apr 15 13:54:04 DESKTOP-72C1QPJ python3[1117]: [2026-04-15 13:54:04.554271] Found 5 SIEM Alerts. Sending to Shuffle...
Apr 15 13:55:05 DESKTOP-72C1QPJ python3[1117]: [2026-04-15 13:55:05.353214] Found 1 SIEM Alerts. Sending to Shuffle...
Apr 15 13:56:05 DESKTOP-72C1QPJ python3[1117]: [2026-04-15 13:56:05.546881] Found 1 SIEM Alerts. Sending to Shuffle...
```

Figure 2: The terminal showing the Python script successfully finding and sending alerts Found X SIEM Alerts

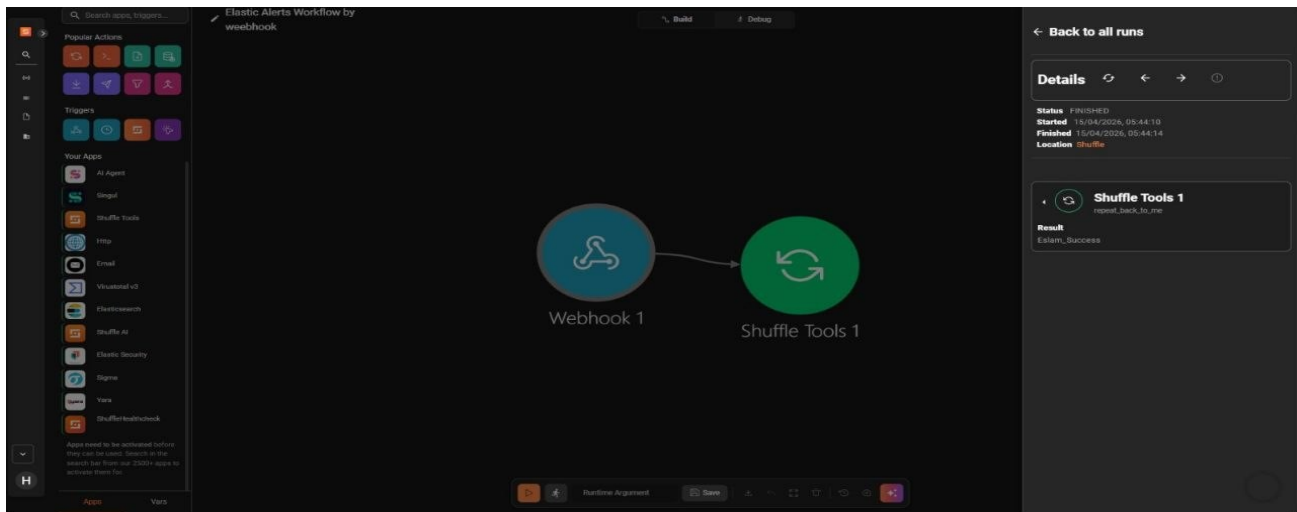


Figure 3 orbis work and shuffle hundle error

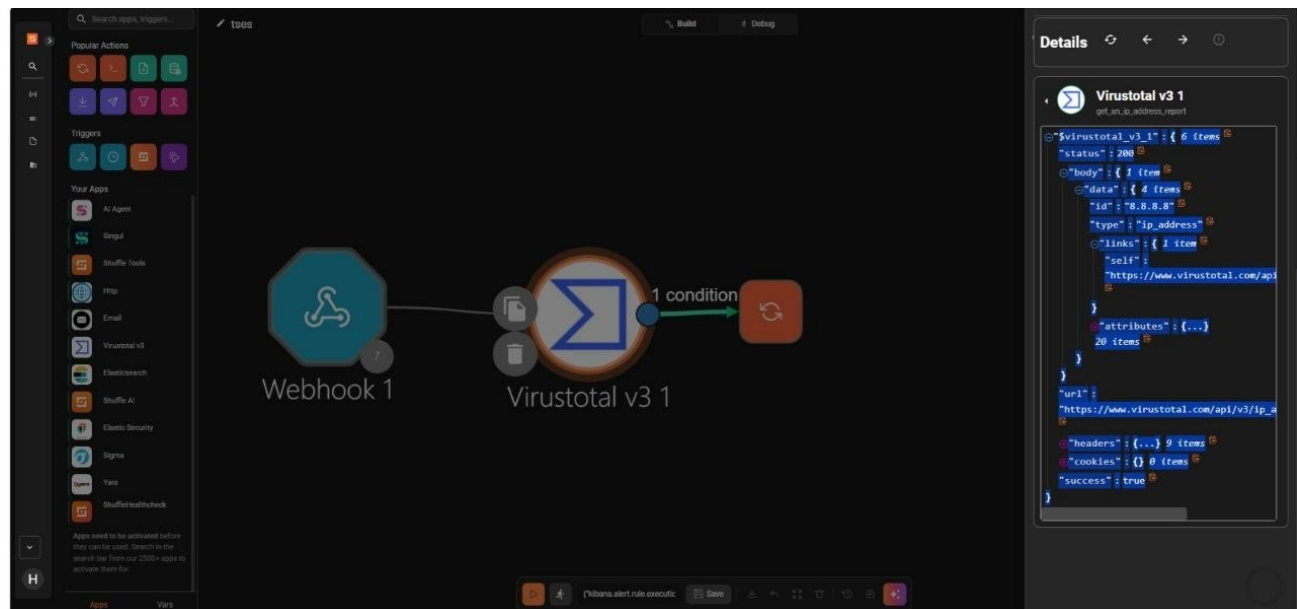


Figure 4: The Shuffle Workflow showing the green execution path through the VirusTotal condition

Figure 4: The terminal showing the Python script successfully finding and sending alerts Found X SIEM Alerts

Figure 5: The Shuffle Workflow showing the green execution path through the VirusTotal condition

Live Product

A fully functional, integrated SIEM-to-SOAR pipeline capable of executing automated playbooks within seconds of a SIEM detection.

Evidence of Achievement

Successful HTTP 200 OK responses from VirusTotal parsing dynamic IPs, and time-stamped logs proving the automated flow of data from detection to notification.

Cyber Kill Chain Mapping

Delivery/Installation: Info Stealer execution.

Command and Control (C2) / Actions on Objectives: Malware attempting to exfiltrate data (Caught by the SIEM rules).

MITRE ATT&CK

TA0001 (Initial Access): Monitored via rules detecting suspicious process executions or unauthorized access.

TA0010 (Exfiltration): Monitored via rules detecting anomalous outbound network traffic to known bad IPs.

5. Executive Summary

This project successfully demonstrated a proactive, automated defense mechanism against Information Stealer malware. By integrating Elastic SIEM with Shuffle SOAR, the detection of Initial Access and Exfiltration behaviors was transformed from a manual monitoring task into a fully automated response pipeline. The system efficiently ingested logs, triggered alerts, and utilized a custom Python bridge to initiate a SOAR playbook. Despite technical challenges involving dynamic JSON data mapping, webhook configurations, and API authentications, the pipeline successfully enriched threat data using VirusTotal and executed automated containment notifications. The results prove the immense value of integrating SOAR into modern SOC environments, paving the way for faster, error-free incident response and elevated network security.